FORM 2
THE PATENT ACT,

1970 (39 of 1970)

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The Patent Rules, 2003

COMPLETE

SPECIFICATION

(See section

10 and rule 13)

1. TITLE OF THE INVENTION:

AI-BASED URBAN GREEN SPACE MONITORING SYSTEM AND BIODIVERSITY ASSESSMENT PORTAL INCORPORATING MULTI-SOURCE GEO-SPATIAL ANALYTICS, ECOLOGICAL INTELLIGENCE, AND PREDICTIVE ENVIRONMENTAL DECISION SUPPORT FRAMEWORK

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3. PREAMBLE TO THE DESCRIPTION:

The following specification particularly describes the invention and the manner in which it is to be performed:

4. DESCRIPTION:

[0001] The present invention relates to an intelligent environmental monitoring system designed to automate the assessment, management, and conservation of urban green spaces through the integration of artificial intelligence, remote sensing technologies, geographic information systems, Internet of Things sensors, and biodiversity analytics. The invention provides a unified digital platform capable of continuously monitoring vegetation health, land cover changes, ecological diversity, environmental quality, and urban ecosystem conditions across parks, forests, wetlands, roadside plantations, institutional campuses, and other green infrastructure. By combining high-resolution satellite imagery, unmanned aerial vehicle observations, sensor-generated environmental data, and AI-based predictive models, the proposed system enables real-time environmental assessment and evidence-based urban ecological management. The platform significantly enhances monitoring accuracy, minimizes manual ecological surveys, supports biodiversity conservation initiatives, and enables proactive decision-making for sustainable urban development.

[0002] The invention further introduces an integrated Biodiversity Assessment Portal capable of collecting, organizing, analyzing, and visualizing ecological information from multiple heterogeneous data sources within a centralized intelligent framework. The portal utilizes advanced machine learning algorithms, deep learning-based image recognition, habitat suitability modeling, species distribution prediction, and ecological risk assessment techniques to automatically identify vegetation species, wildlife occurrences, habitat degradation, invasive species expansion, and ecosystem disturbances. The system continuously evaluates environmental indicators including canopy density, carbon sequestration potential, air quality contribution, soil moisture variability, and habitat connectivity while generating dynamic biodiversity indices and sustainability reports. This intelligent assessment framework enables urban planners, environmental agencies, researchers, conservation organizations, and policymakers to monitor ecosystem health efficiently while improving transparency, scalability, and long-term environmental resilience.

Field of the invention:

[0003] The present invention relates generally to the field of environmental

monitoring systems, geospatial intelligence, and smart urban ecosystem management, particularly focusing on an Artificial Intelligence-based Urban Green Space Monitoring System integrated with a Biodiversity Assessment Portal. The invention combines advanced technologies including artificial intelligence, machine learning, deep learning, Geographic Information Systems (GIS), remote sensing, Internet of Things (IoT), unmanned aerial vehicle (UAV) imaging, and cloud-based environmental analytics for comprehensive monitoring of urban ecological resources. The proposed system enables automated identification, classification, assessment, and prediction of vegetation health, biodiversity richness, habitat quality, and environmental changes occurring within urban landscapes. Furthermore, the invention provides intelligent decision-support capabilities for sustainable urban planning, ecological conservation, climate adaptation, and environmental governance by transforming heterogeneous environmental datasets into meaningful ecological insights. The invention therefore belongs to the interdisciplinary domains of artificial intelligence, environmental informatics, smart city technologies, ecological engineering, and geospatial data science.

[0004] More particularly, the invention pertains to an intelligent multi-source environmental monitoring framework capable of integrating satellite imagery, drone-acquired aerial images, IoT sensor observations, climatic datasets, citizen-contributed biodiversity records, historical land-use information, and spatial ecological databases into a unified analytical platform. The invention employs advanced computer vision algorithms for vegetation recognition, deep learning models for species identification, predictive ecological intelligence for habitat assessment, and geospatial analytics for monitoring land cover transformations and biodiversity dynamics. The system further incorporates automated ecological indexing, habitat suitability modelling, carbon sequestration estimation, urban heat island analysis, pollution impact assessment, and biodiversity risk prediction within a centralized digital portal. Through continuous environmental monitoring and AI-assisted ecological intelligence, the invention enables real-time evaluation of urban ecosystem performance while supporting long-term conservation strategies, regulatory compliance, environmental reporting, and scientifically informed decision-making for sustainable city development.

Background of the invention:

[0005] Rapid urbanization, industrial expansion, and unplanned infrastructure

development have significantly reduced the availability and quality of urban green spaces across cities worldwide. Parks, urban forests, wetlands, roadside plantations, biodiversity corridors, institutional campuses, and community gardens play a critical role in maintaining ecological balance, improving air quality, regulating urban temperature, conserving biodiversity, and enhancing public health. However, conventional methods for monitoring these ecological assets primarily depend on manual field surveys, periodic inspections, and fragmented environmental reports, which are time-consuming, labour-intensive, expensive, and often incapable of providing real-time information. The absence of continuous monitoring mechanisms limits the ability of environmental authorities to detect ecosystem degradation at an early stage, resulting in delayed conservation measures and ineffective resource allocation. Consequently, there exists an increasing demand for intelligent environmental monitoring systems capable of providing automated, scalable, and data-driven ecological assessment.

[0006] Existing urban biodiversity assessment approaches are largely dependent upon ecological experts conducting manual observations of flora and fauna within designated survey regions. Such surveys generally require extensive manpower, specialized taxonomic expertise, repeated field visits, and significant financial investment to generate reliable ecological datasets. Moreover, biodiversity information collected through traditional approaches frequently becomes outdated due to rapid environmental changes caused by urban expansion, climate variability, invasive species, pollution, and anthropogenic disturbances. Since manual observations are often limited in spatial and temporal coverage, many ecological changes remain undetected until substantial habitat degradation has already occurred. These limitations reduce the effectiveness of conservation planning, biodiversity restoration programs, and environmental policy implementation, thereby highlighting the necessity for automated ecological intelligence capable of continuous biodiversity assessment.

[0007] Although satellite remote sensing and drone technologies have considerably improved environmental observation capabilities, most existing monitoring platforms process imagery independently without integrating ecological intelligence, biodiversity analytics, environmental sensor data, or predictive artificial intelligence models. Current systems generally focus only on vegetation mapping or land cover classification while neglecting critical environmental parameters such as habitat

connectivity, species diversity, ecological resilience, carbon storage potential, soil moisture variability, invasive species detection, and ecosystem health indicators. Furthermore, the lack of interoperability between remote sensing platforms, GIS databases, IoT sensor networks, and biodiversity repositories prevents comprehensive ecosystem evaluation. As a result, decision-makers often rely on isolated datasets that fail to provide a holistic understanding of rapidly evolving urban ecological conditions.

[0008] Another significant limitation of conventional environmental management systems is their inability to provide predictive ecological intelligence for future planning and risk mitigation. Most available solutions primarily perform descriptive analysis based on historical observations and are incapable of forecasting vegetation decline, biodiversity loss, habitat fragmentation, urban heat island expansion, pollution impacts, or ecological restoration priorities. Furthermore, these systems rarely employ explainable artificial intelligence or adaptive machine learning models capable of continuously improving prediction accuracy using newly acquired environmental data. The absence of predictive decision-support mechanisms restricts proactive environmental management, making it difficult for urban planners and conservation agencies to prioritize restoration initiatives, allocate resources efficiently, and formulate long-term sustainability strategies.

[0009] Therefore, there exists a critical requirement for an integrated Artificial Intelligence-Based Urban Green Space Monitoring System and Biodiversity Assessment Portal capable of combining multi-source geospatial data, intelligent environmental sensing, automated biodiversity assessment, predictive ecological analytics, and interactive decision-support technologies within a unified platform. Such a system should continuously monitor vegetation health, biodiversity dynamics, ecosystem services, climatic influences, environmental quality, habitat connectivity, and ecological risks while providing accurate, real-time, and scientifically validated information to urban planners, environmental authorities, conservation organizations, researchers, policymakers, and citizens. By overcoming the shortcomings of existing monitoring approaches, the proposed invention aims to establish an intelligent, scalable, transparent, and sustainable framework for preserving urban ecosystems, enhancing biodiversity conservation, improving climate resilience, and supporting evidence-based environmental governance.

Objectives of the invention:

[0010] To develop an Artificial Intelligence-driven Urban Green Space Monitoring System capable of continuously assessing vegetation health, land cover dynamics, and ecological conditions using multi-source geospatial data.

The primary objective of the present invention is to develop an intelligent environmental monitoring system capable of automatically observing and assessing urban green spaces through advanced Artificial Intelligence techniques. The proposed system integrates satellite imagery, drone-acquired aerial photographs, IoT-based environmental sensors, and Geographic Information Systems to generate accurate and real-time ecological information. The invention continuously evaluates vegetation health, canopy density, chlorophyll distribution, land-use variations, and environmental quality without requiring extensive manual intervention. By automating environmental monitoring processes, the system significantly reduces operational costs while increasing monitoring frequency and spatial coverage. Furthermore, the invention enables timely identification of ecological degradation, vegetation stress, and unauthorized land-use modifications, thereby supporting rapid environmental intervention and sustainable urban ecosystem management. The intelligent monitoring framework also improves the reliability, scalability, and consistency of ecological observations across diverse urban landscapes.

[0011] To establish an intelligent Biodiversity Assessment Portal capable of automatically identifying, classifying, and evaluating plant, animal, and microbial diversity using artificial intelligence and computer vision technologies.

Another objective of the present invention is to establish a comprehensive Biodiversity Assessment Portal that enables automated biodiversity documentation and ecological analysis across urban environments. The system employs deep learning models, image recognition algorithms, species classification networks, and ecological databases to accurately identify flora and fauna from satellite images, drone imagery, camera traps, and citizen-contributed observations. The portal continuously evaluates species richness, habitat quality, biodiversity indices, endangered species distribution, invasive species occurrence, and ecological interactions across multiple ecosystems. The invention minimizes dependence on manual ecological surveys while significantly improving the accuracy, consistency, and speed of biodiversity assessment. Additionally, the portal generates dynamic biodiversity inventories, conservation reports, and ecological status dashboards that support researchers,

environmental agencies, conservation authorities, and policymakers in making informed environmental management decisions. This objective promotes long-term biodiversity conservation through continuous intelligent ecological surveillance.

[0012] To integrate multi-source environmental datasets through Geographic Information Systems, remote sensing technologies, Internet of Things devices, and cloud-based data management frameworks for comprehensive ecosystem analysis.

Another objective of the invention is to provide seamless integration of heterogeneous environmental datasets collected from multiple monitoring platforms into a unified analytical framework. The proposed system combines high-resolution satellite imagery, drone-based observations, meteorological information, IoT sensor measurements, soil quality data, hydrological datasets, historical land-use records, and biodiversity databases within a centralized Geographic Information System. The integrated architecture enables synchronized environmental analysis by eliminating data fragmentation commonly observed in existing ecological monitoring systems. Through automated data fusion and spatial intelligence techniques, the invention generates comprehensive environmental maps, ecological indicators, habitat connectivity models, and sustainability assessments. The integrated platform further enhances interoperability among environmental agencies, municipal authorities, researchers, and conservation organizations while supporting standardized environmental reporting and collaborative ecosystem management initiatives. Consequently, the invention establishes a holistic environmental intelligence framework capable of addressing complex urban ecological challenges.

[0013] To implement predictive ecological intelligence capable of forecasting vegetation degradation, biodiversity decline, habitat fragmentation, climate-related risks, and environmental disturbances using machine learning and predictive analytics.

Another objective of the present invention is to incorporate advanced predictive analytics that enable proactive environmental management instead of reactive ecological restoration. The invention utilizes machine learning algorithms, temporal satellite image analysis, climate modelling techniques, environmental trend analysis, and ecological simulation models to forecast future ecosystem changes with high accuracy. The predictive framework estimates vegetation deterioration, biodiversity loss, habitat fragmentation, urban heat island expansion, pollution effects, carbon

sequestration decline, and ecological vulnerability under varying environmental conditions. By generating future environmental scenarios and ecological risk assessments, the system enables authorities to prioritize conservation activities before irreversible environmental damage occurs. The predictive intelligence module continuously improves its forecasting capability through adaptive machine learning models trained using newly acquired environmental datasets. This objective significantly enhances long-term environmental planning, climate resilience, and sustainable urban ecosystem conservation.

[0014] To provide an intelligent environmental decision-support framework capable of assisting urban planners, conservation agencies, governmental organizations, researchers, and policymakers in evidence-based ecological governance.

Another objective of the invention is to establish an AI-powered decision-support system capable of transforming complex environmental information into actionable ecological intelligence for strategic planning and policy formulation. The system generates interactive dashboards, thematic GIS maps, biodiversity scorecards, sustainability indicators, habitat restoration priorities, environmental compliance reports, and ecological performance analytics based on continuously updated environmental observations. The decision-support framework assists municipal corporations, smart city authorities, environmental protection agencies, forestry departments, academic researchers, and conservation organizations in evaluating ecosystem conditions and prioritizing environmental interventions. Furthermore, the invention provides explainable Artificial Intelligence outputs that enhance transparency, accountability, and scientific credibility in ecological decision-making processes. By supporting data-driven governance, the invention contributes significantly toward sustainable urban development, environmental protection, and climate adaptation initiatives.

[0015] To design a scalable, interoperable, and adaptive environmental monitoring platform capable of supporting smart city ecosystems, biodiversity conservation programs, climate resilience initiatives, and sustainable urban development across diverse geographical regions.

Another objective of the present invention is to develop a highly scalable and adaptable environmental monitoring platform capable of deployment across cities, metropolitan regions, industrial zones, educational campuses, protected forests,

biodiversity parks, wetlands, and other ecologically significant landscapes. The proposed architecture supports cloud computing infrastructure, distributed environmental databases, interoperable GIS platforms, standardized environmental data exchange protocols, and modular Artificial Intelligence services to ensure seamless expansion and long-term operational sustainability. The invention accommodates increasing volumes of satellite imagery, IoT sensor streams, biodiversity observations, and environmental records without compromising analytical performance or system responsiveness. Moreover, the modular design enables future integration of emerging technologies including autonomous environmental robots, advanced hyperspectral imaging, digital twins, climate intelligence models, and next-generation ecological sensors. This objective ensures that the invention remains technologically relevant, environmentally sustainable, and operationally efficient for future smart city and environmental conservation ecosystems.

Summary of the invention:

[0016] Artificial Intelligence-driven Environmental Intelligence Framework for Comprehensive Urban Green Space Monitoring and Biodiversity Assessment using Multi-Source Geospatial Analytics

The present invention introduces an integrated Artificial Intelligence-based environmental intelligence framework designed to continuously monitor, evaluate, and manage urban green spaces through advanced computational technologies. The framework combines satellite remote sensing, unmanned aerial vehicle imaging, Geographic Information Systems, Internet of Things environmental sensors, and cloud computing infrastructure within a unified ecological monitoring platform. The system automatically acquires, preprocesses, and analyses heterogeneous environmental datasets to generate accurate information regarding vegetation health, canopy density, land-use dynamics, biodiversity richness, ecological resilience, and habitat quality. By replacing conventional manual monitoring approaches with intelligent automation, the invention significantly improves monitoring accuracy, operational efficiency, and temporal consistency. The framework further supports continuous environmental surveillance, rapid anomaly detection, and real-time ecological reporting for sustainable urban ecosystem management.

[0017] Integrated Multi-Source Data Fusion Architecture for Real-Time Ecological Monitoring, Spatial Intelligence, and Environmental Decision Support

The invention employs a comprehensive multi-source environmental data fusion architecture capable of integrating information collected from high-resolution satellite imagery, drone-acquired aerial photographs, IoT sensor networks, meteorological stations, biodiversity repositories, land-use databases, and historical ecological records. The integrated architecture performs automated spatial alignment, data normalization, feature extraction, and temporal synchronization to eliminate inconsistencies among heterogeneous environmental datasets. Advanced Geographic Information System technologies generate dynamic ecological maps, spatial distribution models, habitat connectivity analyses, and environmental change detection outputs that provide a holistic understanding of urban ecosystem dynamics. Furthermore, the architecture supports seamless interoperability between municipal authorities, environmental agencies, conservation organizations, academic researchers, and urban planners by enabling standardized environmental information exchange.

[0018] Artificial Intelligence and Deep Learning-Based Biodiversity Identification, Ecological Classification, and Habitat Assessment Framework.

The invention incorporates advanced Artificial Intelligence and deep learning models capable of automatically identifying, classifying, and evaluating plant species, wildlife populations, ecological habitats, invasive organisms, and biodiversity indicators using high-resolution environmental imagery and ecological datasets. Computer vision algorithms analyse drone photographs, satellite images, camera trap observations, and citizen-generated biodiversity records to detect vegetation types, species distributions, habitat fragmentation, and ecological disturbances with high precision. The system continuously calculates biodiversity indices, habitat suitability scores, conservation priorities, species richness, ecological diversity metrics, and environmental sustainability indicators using adaptive machine learning algorithms. Furthermore, intelligent ecological classification enables automatic recognition of endangered species habitats, biodiversity hotspots, degraded ecosystems, and restoration opportunities requiring immediate environmental intervention.

[0019] Predictive Environmental Analytics Engine Utilizing Machine Learning for Ecological Forecasting, Climate Risk Assessment, and Sustainable Urban Planning

The invention further introduces an intelligent predictive analytics engine capable of forecasting future ecological conditions through advanced machine learning, temporal environmental modelling, and climate-sensitive analytical techniques. The predictive engine continuously analyses historical environmental observations together with real-time ecological data to estimate vegetation degradation, biodiversity decline, habitat fragmentation, pollution intensity, urban heat island expansion, flood susceptibility, drought impacts, and carbon sequestration variability. By identifying environmental risks before irreversible ecological degradation occurs, the invention enables proactive conservation planning and optimized allocation of environmental restoration resources. Adaptive Artificial Intelligence models continuously improve prediction accuracy through incremental learning from newly acquired environmental observations and changing climatic conditions.

[0020] Cloud-Based Environmental Decision Support Portal Incorporating Interactive GIS Visualization, Automated Sustainability Reporting, and Explainable Artificial Intelligence.

The invention provides a cloud-enabled Biodiversity Assessment Portal designed to deliver intelligent environmental decision-support services through interactive visualization technologies, explainable Artificial Intelligence models, and comprehensive ecological reporting mechanisms. The portal generates dynamic Geographic Information System dashboards, biodiversity scorecards, vegetation health reports, carbon sequestration estimates, environmental compliance assessments, habitat restoration recommendations, pollution impact analyses, and sustainability performance indicators using continuously updated environmental observations. Explainable Artificial Intelligence modules present transparent reasoning behind ecological predictions, species classifications, conservation priorities, and environmental risk assessments, thereby improving stakeholder confidence and regulatory accountability. The centralized portal further facilitates secure information sharing among governmental agencies, municipal corporations, conservation authorities, academic institutions, environmental researchers, and policy-makers through cloud-based collaborative infrastructure.

[0021] Scalable Smart City Environmental Monitoring Ecosystem Supporting Biodiversity Conservation, Climate Resilience, Regulatory Compliance, and Long-Term Sustainable Urban Development

The invention establishes a scalable environmental monitoring ecosystem specifically designed to support emerging smart city initiatives, biodiversity conservation programs, climate resilience strategies, environmental governance frameworks, and sustainable urban infrastructure development. The modular architecture accommodates continuous expansion through integration of additional remote sensing platforms, autonomous environmental monitoring devices, advanced ecological sensors, hyperspectral imaging systems, digital twin technologies, and next-generation Artificial Intelligence services without requiring significant system redesign. The platform efficiently manages large-scale environmental datasets collected across multiple cities, protected forests, wetlands, institutional campuses, industrial green belts, urban biodiversity parks, and ecological restoration sites while maintaining high computational performance and analytical reliability. Moreover, standardized environmental data models and interoperable system architecture ensure compatibility with national environmental information systems, international biodiversity databases, and future ecological monitoring technologies.

Detailed description of the drawings

[0022] Figure 1 – Overall Workflow of the AI-Based Urban Green Space Monitoring System and Biodiversity Assessment Portal

Figure 1 illustrates the complete operational workflow of the proposed Artificial Intelligence-Based Urban Green Space Monitoring System and Biodiversity Assessment Portal. The workflow begins with the acquisition of environmental information from multiple heterogeneous sources including high-resolution satellite imagery, unmanned aerial vehicle (UAV) images, Internet of Things environmental sensors, meteorological stations, GIS databases, and biodiversity observation records. The collected datasets undergo preprocessing operations comprising image enhancement, geo-referencing, data normalization, noise removal, spatial alignment, and feature extraction before entering the Artificial Intelligence processing engine. The AI engine subsequently performs vegetation classification, biodiversity assessment, habitat evaluation, environmental quality analysis, land-use change detection, and ecological health estimation using deep learning and machine learning algorithms. The generated ecological information is stored within a centralized environmental repository and visualized through the Biodiversity Assessment Portal using interactive GIS dashboards, thematic maps, analytical reports, and ecological scorecards. The workflow concludes with intelligent environmental recommendations, sustainability

reports, conservation priorities, and predictive ecological alerts delivered to urban planners, environmental agencies, researchers, and policymakers for evidence-based environmental governance.

[0023] Figure 2 – Technical Architecture of the Artificial Intelligence-Based Urban Green Space Monitoring and Biodiversity Intelligence Framework

Figure 2 depicts the comprehensive technical architecture of the proposed invention, illustrating the interaction among data acquisition modules, cloud infrastructure, Artificial Intelligence services, geospatial processing engines, biodiversity analytics modules, and user interface components. The architecture comprises multiple environmental data collection sources including satellite imagery services, drone imaging platforms, IoT sensor gateways, weather stations, biodiversity databases, and GIS repositories connected through secure application programming interfaces. The acquired environmental information is transmitted to a cloud-based processing infrastructure where preprocessing, feature extraction, spatial indexing, and environmental data fusion are performed. The Artificial Intelligence engine incorporates computer vision models, deep learning classifiers, ecological prediction modules, habitat assessment algorithms, and biodiversity identification networks coordinated through an intelligent orchestration framework. The processed information is subsequently stored within centralized environmental databases and delivered to the Biodiversity Assessment Portal for visualization, reporting, predictive analytics, sustainability assessment, and environmental decision support. The modular architecture ensures scalability, interoperability, high computational efficiency, and seamless integration with future environmental monitoring technologies.

[0024] Figure 3 – Intelligent Ecological Assessment and Predictive Environmental Decision-Support Process

Figure 3 presents the intelligent ecological assessment and predictive environmental decision-support process implemented within the proposed invention. The process begins with automated environmental feature extraction from satellite imagery, drone observations, IoT sensor measurements, and biodiversity records to identify vegetation characteristics, habitat conditions, climatic variables, and ecological indicators. The extracted environmental features are analysed using Artificial Intelligence models capable of vegetation health estimation, biodiversity classification, habitat suitability evaluation, carbon sequestration assessment, pollution analysis, and ecological resilience measurement. The resulting

environmental intelligence is subsequently processed through predictive machine learning models that forecast vegetation degradation, biodiversity decline, habitat fragmentation, urban heat island expansion, flood vulnerability, drought susceptibility, and restoration priorities under varying environmental scenarios. The generated predictions are further integrated with Geographic Information System visualization modules that produce ecological risk maps, biodiversity distribution models, conservation priority zones, and sustainability dashboards. Finally, the system generates explainable Artificial Intelligence recommendations, environmental compliance reports, restoration strategies, and decision-support notifications that assist governmental agencies, conservation authorities, municipal corporations, researchers, and policymakers in implementing effective ecosystem conservation and sustainable urban planning initiatives.

Detailed invention disclosure:

[0025] Multi-Source Environmental Data Acquisition and Intelligent Preprocessing Module

The invention provides an intelligent environmental data acquisition mechanism capable of collecting heterogeneous ecological information from multiple monitoring sources including high-resolution satellite imagery, unmanned aerial vehicles, Internet of Things environmental sensors, weather stations, Geographic Information System databases, biodiversity repositories, and citizen-generated ecological observations. The acquired environmental datasets are automatically validated for completeness, spatial consistency, and temporal synchronization before entering the preprocessing pipeline. The preprocessing module performs image enhancement, atmospheric correction, geo-referencing, coordinate transformation, data normalization, missing value estimation, noise reduction, and environmental metadata extraction to improve analytical accuracy. Furthermore, spatial alignment algorithms synchronize datasets originating from different resolutions and acquisition frequencies into a unified environmental reference framework. The standardized environmental information is subsequently prepared for Artificial Intelligence-based analysis while preserving ecological accuracy and spatial integrity.

[0026] Artificial Intelligence-Based Vegetation Detection and Urban Green Space Classification Module

The invention incorporates an advanced Artificial Intelligence-based vegetation

detection module configured to automatically identify, classify, and monitor various categories of urban green spaces using high-resolution satellite imagery, unmanned aerial vehicle images, multispectral datasets, and Geographic Information System layers. The module utilizes deep convolutional neural networks, semantic image segmentation algorithms, and machine learning classifiers to distinguish parks, urban forests, roadside plantations, wetlands, recreational gardens, institutional landscapes, green rooftops, ecological corridors, and other vegetation-covered regions from surrounding built-up infrastructure. The identified vegetation classes are continuously evaluated for canopy density, vegetation coverage percentage, Normalized Difference Vegetation Index (NDVI), Enhanced Vegetation Index (EVI), and chlorophyll concentration to determine ecosystem health. Furthermore, the module automatically detects unauthorized vegetation removal, illegal encroachments, deforestation activities, and land-use modifications occurring within monitored regions. The classified green space information is periodically updated within the centralized geospatial database, enabling authorities to maintain an accurate digital inventory of urban ecological assets.

[0027] Automated Biodiversity Identification and Ecological Species Recognition Framework

The invention further provides an intelligent biodiversity identification framework capable of automatically recognizing and classifying flora, fauna, avian species, insects, amphibians, and other ecological organisms using Artificial Intelligence and computer vision technologies. The framework analyses environmental images captured through drones, camera traps, mobile devices, satellite observations, and citizen science applications to detect species-specific morphological characteristics and ecological attributes. Deep learning classification models compare extracted features against standardized biodiversity databases and taxonomic repositories to accurately identify species, determine conservation status, and estimate species abundance within monitored ecosystems. Additionally, the framework identifies endangered species, invasive organisms, migratory wildlife, and ecologically sensitive habitats requiring immediate conservation attention. Continuous biodiversity assessment enables automatic generation of species richness indices, ecological diversity metrics, habitat occupancy maps, and biodiversity trend analyses across different temporal intervals.

[0028] Habitat Suitability Assessment and Ecological Health Evaluation Engine

The invention includes a comprehensive habitat suitability assessment engine configured to evaluate the ecological quality, sustainability, and resilience of urban habitats using integrated environmental intelligence. The engine analyses vegetation density, soil moisture content, climatic variables, water availability, habitat connectivity, pollution intensity, anthropogenic disturbances, and biodiversity distribution to determine the ecological suitability of different geographical regions for sustaining diverse biological communities. Artificial Intelligence models calculate habitat quality indices, ecological integrity scores, environmental carrying capacity, restoration priorities, and ecosystem stability metrics using adaptive machine learning techniques. Furthermore, the system continuously monitors habitat degradation caused by urban expansion, industrial development, climate variability, invasive species, and environmental pollution while generating early warning indicators for declining ecological conditions.

[0029] Urban Land Use Change Detection and Green Space Transformation Analysis Module

The invention incorporates an intelligent land-use change detection module capable of continuously monitoring temporal transformations occurring within urban green spaces through comparative geospatial analysis of historical and real-time environmental datasets. The module utilizes satellite image differencing, deep learning-based object detection, temporal GIS modelling, and change vector analysis techniques to identify vegetation loss, urban expansion, construction activities, land conversion, ecological fragmentation, and restoration initiatives. Artificial Intelligence algorithms automatically classify detected land-use changes according to ecological significance, environmental impact, and conservation priority while generating accurate spatial change maps and transformation reports. The system additionally quantifies changes in green cover percentage, tree canopy distribution, open ecological spaces, wetland boundaries, and habitat continuity across multiple observation periods. Automated notifications are generated whenever environmentally sensitive areas experience unauthorized development or significant ecological degradation.

[0030] Environmental Quality Monitoring and Climate Resilience Assessment Framework

The invention further provides an intelligent environmental quality monitoring

framework designed to continuously evaluate atmospheric, climatic, hydrological, and ecological conditions influencing urban green spaces and biodiversity. The framework integrates environmental observations collected from IoT sensors, weather stations, air quality monitoring devices, rainfall measurement systems, soil moisture sensors, groundwater monitoring stations, and climatic databases to generate comprehensive environmental intelligence. Artificial Intelligence models analyse environmental parameters including temperature fluctuations, humidity, precipitation, particulate matter concentration, carbon dioxide levels, soil fertility, water availability, and pollution intensity to determine ecosystem health and climate resilience. The system further estimates urban heat island effects, drought susceptibility, flood vulnerability, carbon sequestration potential, and ecological stress levels using predictive environmental analytics. Continuous environmental quality monitoring enables timely identification of climate-induced ecological disturbances while supporting adaptive conservation planning and sustainable urban ecosystem management.

[0031] Internet of Things-Based Real-Time Environmental Sensor Integration and Continuous Ecosystem Monitoring Module

The invention incorporates an Internet of Things (IoT)-based environmental monitoring module configured to continuously collect real-time ecological information from distributed sensor networks deployed across urban green spaces. The sensor network comprises air quality sensors, temperature and humidity sensors, soil moisture probes, light intensity sensors, rainfall gauges, groundwater monitoring devices, wind sensors, and acoustic biodiversity monitoring equipment that periodically transmit environmental observations to the centralized cloud platform. The acquired sensor information is automatically synchronized with satellite imagery, drone observations, and Geographic Information System datasets to establish a comprehensive real-time environmental intelligence framework. Artificial Intelligence algorithms continuously analyse sensor streams to detect abnormal ecological conditions including water stress, excessive pollution, declining soil fertility, habitat disturbances, abnormal climatic variations, and vegetation deterioration. The module further generates automated environmental alerts whenever predefined ecological thresholds are exceeded, thereby enabling immediate corrective action by environmental authorities.

[0032] Artificial Intelligence-Based Ecological Risk Prediction and

Environmental Early Warning System

The invention provides an intelligent ecological risk prediction engine configured to forecast environmental threats before they result in irreversible ecosystem degradation. The predictive framework utilizes machine learning algorithms, temporal environmental modelling, historical ecological records, climatic datasets, biodiversity observations, and geospatial intelligence to estimate future ecological conditions with high accuracy. The system continuously evaluates environmental parameters associated with vegetation decline, biodiversity loss, habitat fragmentation, invasive species expansion, drought occurrence, flood susceptibility, urban heat island intensification, wildfire vulnerability, and pollution accumulation. Based on predictive analysis, the engine assigns environmental risk scores to monitored locations and categorizes ecological regions according to conservation urgency and restoration priority. Additionally, the system generates automated early warning notifications, ecological vulnerability reports, and predictive GIS risk maps for urban planners, environmental agencies, disaster management authorities, and conservation organizations.

[0033] Carbon Sequestration Estimation and Ecosystem Service Assessment Framework

The invention further includes an intelligent ecosystem service assessment framework capable of estimating the environmental benefits provided by urban green spaces using Artificial Intelligence and geospatial analytics. The framework automatically calculates carbon sequestration potential, oxygen generation capacity, air pollution mitigation efficiency, stormwater retention capability, soil conservation performance, urban temperature regulation, and ecological service values based on vegetation characteristics, canopy density, biomass estimation, climatic conditions, and land-use information. Advanced machine learning models evaluate ecological productivity by integrating multispectral satellite imagery, LiDAR-derived vegetation measurements, and field-based environmental observations. The system periodically generates carbon inventory reports, ecosystem valuation metrics, sustainability indices, and climate mitigation assessments that support national environmental reporting and carbon neutrality initiatives. Furthermore, ecological service assessments assist policymakers in quantifying the environmental contributions of urban green infrastructure for informed investment and conservation planning.

[0034] Cloud-Based Biodiversity Assessment Portal with Interactive

Geographic Information System Visualization

The invention provides a centralized cloud-based Biodiversity Assessment Portal configured to facilitate secure storage, visualization, analysis, and dissemination of ecological intelligence through interactive Geographic Information System interfaces. The portal integrates processed environmental information including vegetation distribution, biodiversity observations, habitat quality assessments, environmental sensor data, predictive ecological models, conservation priorities, and sustainability indicators into a unified digital platform accessible to authorized stakeholders. Interactive dashboards present dynamic GIS maps, ecological heat maps, biodiversity indices, temporal vegetation trends, species distribution layers, environmental quality reports, and restoration progress through intuitive visualization tools. The portal further enables customized environmental reporting, spatial querying, biodiversity inventory management, ecological comparison analysis, and downloadable sustainability documentation according to user-defined geographical regions and monitoring periods. Secure cloud infrastructure ensures reliable data accessibility, scalability, disaster recovery, and collaborative information sharing among governmental agencies, research institutions, municipal corporations, conservation organizations, and academic communities.

[0035] Automated Environmental Recommendation and Intelligent Conservation Planning Module

The invention incorporates an Artificial Intelligence-driven recommendation engine configured to generate scientifically optimized environmental conservation strategies based on continuously monitored ecological conditions and predictive environmental intelligence. The recommendation module analyses vegetation health, biodiversity richness, habitat quality, environmental risks, climate resilience indicators, pollution levels, ecological service values, and restoration priorities to formulate adaptive conservation plans for individual urban ecosystems. Artificial Intelligence algorithms recommend tree plantation programs, habitat restoration measures, invasive species control strategies, water conservation initiatives, pollution mitigation techniques, biodiversity enhancement projects, and sustainable landscape management practices tailored to specific environmental conditions. Furthermore, the module automatically prioritizes ecological intervention zones according to environmental sensitivity, biodiversity significance, restoration feasibility, and anticipated ecological benefits while optimizing resource allocation for conservation authorities. Explainable Artificial Intelligence mechanisms provide transparent reasoning behind every

recommendation, thereby enhancing stakeholder confidence and regulatory accountability.

[0036] System Integration, Adaptive Learning Framework, and Continuous Environmental Intelligence Optimization

The invention finally incorporates an adaptive learning and system integration framework designed to ensure continuous improvement of environmental monitoring accuracy, predictive capability, and ecological decision-support performance throughout long-term deployment. The framework continuously acquires newly generated satellite imagery, biodiversity observations, environmental sensor measurements, climate records, user feedback, and ecological validation data to retrain Artificial Intelligence models using adaptive machine learning techniques. Performance evaluation mechanisms periodically assess prediction accuracy, species classification efficiency, vegetation detection precision, habitat assessment reliability, and environmental recommendation effectiveness using standardized ecological performance indicators. The modular architecture further supports seamless integration with emerging technologies including hyperspectral satellite systems, autonomous environmental monitoring robots, digital twin ecosystems, advanced LiDAR platforms, blockchain-enabled environmental auditing, and future smart city infrastructures without requiring substantial architectural modifications.

Example 1: Automated Urban Park Monitoring and Vegetation Health Assessment

Method of Procedure

The invention monitors an urban park using satellite imagery, drone images, IoT sensors, weather data, and GIS databases. The collected data is preprocessed and analyzed using AI to assess vegetation health, canopy density, NDVI, chlorophyll levels, biomass, and tree conditions. The system detects vegetation stress, disease, unauthorized tree removal, and declining green cover. The Biodiversity Assessment Portal generates dashboards, vegetation reports, conservation recommendations, and restoration priority maps for efficient park management.

Observation

The system enabled continuous and accurate environmental monitoring with minimal manual intervention. AI-based analysis rapidly detected degraded vegetation, generated reliable ecological reports, and supported timely restoration planning. The invention significantly improved monitoring efficiency, ecological intelligence, and

sustainable urban green space management.

Example 2: Intelligent Biodiversity Assessment and Ecological Risk Prediction in an Urban Wetland Ecosystem

Method of Procedure

The invention is deployed in an urban wetland using drone images, camera traps, IoT water sensors, satellite imagery, biodiversity databases, and climate records. AI models automatically identify species, evaluate habitat quality, and analyse ecological conditions. Predictive machine learning forecasts biodiversity decline, habitat degradation, pollution impacts, and climate risks. The portal generates habitat suitability maps, biodiversity reports, ecological risk maps, and conservation recommendations.

Observation

The system accurately identified species and predicted ecological risks while reducing reliance on manual biodiversity surveys. Interactive GIS dashboards improved environmental visualization and conservation planning. The invention enhanced biodiversity monitoring, ecological forecasting, and sustainable wetland ecosystem management.

ADVANTAGES OF THE AI-BASED URBAN GREEN SPACE MONITORING SYSTEM AND BIODIVERSITY ASSESSMENT PORTAL

[0037] Comprehensive Real-Time Monitoring of Urban Green Spaces through Artificial Intelligence, Remote Sensing, and Multi-Source Environmental Data Integration

The invention enables continuous real-time monitoring of urban green spaces by integrating Artificial Intelligence, Geographic Information Systems, satellite imagery, drone observations, Internet of Things sensors, and environmental databases within a unified intelligent monitoring framework. Unlike conventional ecological surveys that rely on periodic manual inspections, the proposed system continuously observes vegetation health, biodiversity conditions, habitat quality, and environmental changes across large geographical regions. The intelligent monitoring framework significantly improves spatial coverage, temporal resolution, and analytical consistency while minimizing operational delays and human intervention. Automated environmental intelligence enables rapid identification of ecological degradation, unauthorized land-use changes, vegetation stress, and biodiversity decline before irreversible

environmental damage occurs. Consequently, the invention substantially enhances urban environmental management through timely ecological assessment and continuous ecosystem surveillance.

[0038] Improved Accuracy in Biodiversity Identification and Ecological Assessment through Deep Learning and Computer Vision Technologies

The proposed invention significantly improves the precision and reliability of biodiversity assessment by employing advanced deep learning algorithms, computer vision techniques, and Artificial Intelligence-based ecological classification models. Automated species identification minimizes errors commonly associated with manual ecological surveys while enabling rapid recognition of flora, fauna, invasive organisms, endangered species, and habitat characteristics. The intelligent assessment framework continuously updates biodiversity inventories using newly acquired environmental observations, thereby ensuring current and scientifically accurate ecological information. Furthermore, adaptive machine learning algorithms improve classification performance over time by learning from validated environmental datasets. This enhanced analytical capability supports reliable biodiversity conservation planning, scientific ecological research, and long-term environmental sustainability.

[0039] Enhanced Environmental Decision Support through Predictive Ecological Intelligence and Explainable Artificial Intelligence

The invention provides intelligent environmental decision-support capabilities by integrating predictive machine learning models with explainable Artificial Intelligence techniques. The system forecasts vegetation degradation, habitat fragmentation, biodiversity decline, climate-related ecological risks, pollution impacts, and restoration priorities before environmental degradation becomes irreversible. Explainable Artificial Intelligence enables stakeholders to understand the reasoning behind every ecological prediction, conservation recommendation, and environmental risk assessment, thereby improving transparency, accountability, and regulatory acceptance. Interactive dashboards, GIS visualization tools, and sustainability reports assist urban planners, conservation agencies, municipal authorities, and policymakers in making informed environmental decisions based on continuously updated ecological intelligence. Consequently, the invention transforms complex environmental data into practical and actionable conservation strategies.

[0040] Reduced Operational Cost and Human Dependency through Intelligent Environmental Automation

The invention substantially reduces operational expenditure associated with conventional environmental monitoring by automating repetitive ecological assessment activities including vegetation classification, biodiversity identification, habitat evaluation, environmental reporting, and ecological risk analysis. Automated environmental intelligence minimizes dependence on extensive field surveys, manual biodiversity documentation, and repetitive environmental inspections, thereby reducing labour requirements and improving monitoring efficiency. Artificial Intelligence-based automation enables simultaneous assessment of multiple geographical regions with greater consistency and significantly shorter processing time compared to traditional ecological methodologies. The reduction in manual intervention also minimizes human error, improves analytical reproducibility, and ensures standardized environmental reporting across different administrative regions. Consequently, environmental organizations can allocate resources more efficiently while maintaining higher ecological monitoring performance.

[0041] High Scalability, Interoperability, and Future Technology Integration for Smart City Environmental Management.

The modular system architecture enables seamless scalability across municipalities, metropolitan regions, biodiversity parks, forests, wetlands, industrial green belts, educational campuses, and national environmental monitoring programs. The invention supports interoperability with Geographic Information Systems, remote sensing platforms, environmental databases, smart city infrastructure, climate monitoring systems, and governmental ecological information networks through standardized communication protocols and cloud-based integration services. Furthermore, the adaptive architecture allows future incorporation of hyperspectral imaging technologies, autonomous environmental robots, digital twin ecosystems, blockchain-based environmental auditing, advanced LiDAR platforms, and next-generation Artificial Intelligence services without requiring significant structural modifications. This flexibility ensures long-term technological relevance and sustainable deployment across diverse environmental management applications.

[0042] Promotion of Sustainable Urban Development, Climate Resilience, and Long-Term Biodiversity Conservation

The invention contributes significantly toward sustainable urban development by

enabling intelligent conservation of green infrastructure, enhancement of biodiversity, optimization of ecosystem services, and improvement of urban climate resilience. Continuous ecological monitoring assists governmental agencies in preserving natural habitats, improving carbon sequestration, mitigating urban heat island effects, enhancing air quality, conserving water resources, and restoring environmentally degraded ecosystems. Predictive ecological intelligence further supports proactive conservation planning by identifying high-priority restoration areas before severe ecological deterioration occurs. The generated sustainability reports and biodiversity indicators facilitate compliance with environmental regulations, national biodiversity strategies, climate adaptation policies, and international sustainable development objectives. Consequently, the invention establishes a comprehensive technological framework for achieving environmentally resilient, biodiversity-rich, and sustainable smart cities.

5. CLAIMS:

I/We Claim:

[0043] An Artificial Intelligence-Based Urban Green Space Monitoring System and Biodiversity Assessment Portal, comprising: an environmental data acquisition module configured to collect ecological data from satellite imagery, UAV imagery, GIS databases, IoT sensors, meteorological stations, biodiversity repositories, and citizen observations; a preprocessing engine configured to perform data enhancement, geo-referencing, normalization, data fusion, feature extraction, synchronization, and validation; an Artificial Intelligence processing engine comprising machine learning, deep learning, computer vision, and geospatial intelligence modules configured to classify vegetation, identify biodiversity, assess habitat quality, evaluate vegetation health, detect ecological changes, and generate environmental intelligence; a predictive analytics engine configured to forecast vegetation degradation, biodiversity loss, habitat fragmentation, pollution risks, and climate-related ecological changes; a cloud-based Biodiversity Assessment Portal configured to display GIS dashboards, biodiversity inventories, environmental reports, sustainability indicators, and predictive risk maps; and an intelligent recommendation module configured to generate restoration strategies, biodiversity conservation plans, climate adaptation measures, and sustainable urban management recommendations, wherein the system continuously integrates multi-source environmental data, performs automated ecological assessment, enables real-time monitoring, predictive decision support, and intelligent urban ecosystem

management with minimal manual intervention.

[0044] The system as claimed in Claim 1, wherein the environmental data acquisition module is configured to integrate satellite imagery, drone-acquired aerial photographs, multispectral images, LiDAR datasets, Internet of Things sensor observations, meteorological information, biodiversity records, and Geographic Information System layers into a unified environmental intelligence framework, thereby enabling continuous and synchronized monitoring of vegetation health, biodiversity distribution, habitat conditions, climatic variations, and ecological changes across multiple urban environments with improved spatial accuracy, temporal consistency, and analytical reliability.

[0045] The system as claimed in Claim 1, wherein the Artificial Intelligence processing engine comprises deep convolutional neural networks, semantic segmentation models, computer vision algorithms, species recognition networks, and adaptive machine learning classifiers configured to automatically identify vegetation categories, classify green spaces, recognize flora and fauna, estimate canopy density, calculate vegetation indices, evaluate habitat quality, detect invasive species, identify biodiversity hotspots, and continuously improve analytical performance through adaptive learning from newly acquired environmental datasets.

[0046] The system as claimed in Claim 1, wherein the predictive ecological analytics engine is configured to analyze historical environmental observations, climatic datasets, biodiversity inventories, vegetation characteristics, pollution indicators, land-use changes, and environmental sensor measurements to generate predictive models capable of forecasting vegetation degradation, biodiversity decline, habitat fragmentation, ecological vulnerability, drought conditions, flood susceptibility, urban heat island effects, and conservation priorities using explainable Artificial Intelligence and machine learning algorithms.

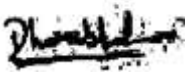
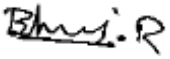
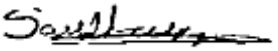
[0047] The system as claimed in Claim 1, wherein the Biodiversity Assessment Portal is configured to provide cloud-based environmental visualization through interactive Geographic Information System dashboards, ecological heat maps, biodiversity scorecards, species distribution models, sustainability reports, environmental compliance assessments, restoration priority maps, conservation analytics, and downloadable ecological documentation while enabling secure information sharing

among governmental agencies, municipal authorities, research organizations, environmental experts, and policy-makers through authenticated cloud infrastructure.

[0048] The system as claimed in Claim 1, wherein the intelligent environmental recommendation module is configured to generate automated biodiversity conservation strategies, ecological restoration plans, afforestation recommendations, invasive species management guidelines, climate resilience measures, pollution mitigation strategies, habitat enhancement programs, carbon sequestration optimization plans, and sustainable urban landscape management recommendations by analyzing continuously updated environmental intelligence and predictive ecological assessment results, thereby supporting evidence-based environmental governance and long-term sustainable urban development.

6. DATE AND SIGNATURE:

Dated 7th August, 2026

Mrs. Saranya N | Dharnish J | Bharathwaj R | Sasidharan R

Name and Signature of the Applicants

ABSTRACT:

Title: AI-BASED URBAN GREEN SPACE MONITORING SYSTEM AND BIODIVERSITY ASSESSMENT PORTAL INCORPORATING MULTI-SOURCE GEO-SPATIAL ANALYTICS, ECOLOGICAL INTELLIGENCE, AND PREDICTIVE ENVIRONMENTAL DECISION SUPPORT FRAMEWORK

Urban green spaces play a vital role in maintaining ecological balance, supporting biodiversity, mitigating climate change, improving air quality, regulating urban temperatures, and enhancing the quality of life within rapidly expanding cities. Conventional environmental monitoring approaches primarily rely on manual field surveys, fragmented ecological databases, and periodic inspections, which are often labour-intensive, time-consuming, expensive, and incapable of providing continuous real-time environmental intelligence. The absence of integrated monitoring frameworks limits the ability of environmental authorities to detect ecological degradation, biodiversity decline, habitat fragmentation, and environmental risks at an early stage, thereby reducing the effectiveness of conservation initiatives and sustainable urban planning.

The present invention proposes an **Artificial Intelligence-Based Urban Green Space Monitoring System and Biodiversity Assessment Portal** that integrates high-resolution satellite imagery, unmanned aerial vehicle observations, Geographic Information Systems, Internet of Things environmental sensors, meteorological datasets, biodiversity repositories, and cloud computing technologies into a unified environmental intelligence platform. The invention employs advanced Artificial Intelligence, deep learning, computer vision, and machine learning algorithms to automatically identify vegetation, classify biodiversity, evaluate habitat quality, estimate vegetation health, monitor ecological changes, assess ecosystem services, and continuously generate environmental intelligence. The integrated Biodiversity Assessment Portal provides interactive Geographic Information System dashboards, biodiversity inventories, ecological reports, predictive environmental analytics, conservation recommendations, sustainability indicators, and automated decision-support services for governmental agencies, researchers, urban planners, and environmental organizations.

The invention further incorporates predictive ecological intelligence capable of forecasting vegetation degradation, biodiversity loss, habitat fragmentation, pollution impacts, urban heat island expansion, climate-related environmental risks, and ecosystem resilience using adaptive machine learning models. Explainable Artificial

Intelligence techniques enhance transparency in ecological predictions while enabling evidence-based conservation planning and environmental governance. By automating ecological assessment, integrating heterogeneous environmental datasets, improving biodiversity monitoring accuracy, reducing manual intervention, and supporting sustainable urban ecosystem management, the proposed invention establishes a scalable, intelligent, and future-ready framework for biodiversity conservation, climate resilience, environmental sustainability, and smart city development.

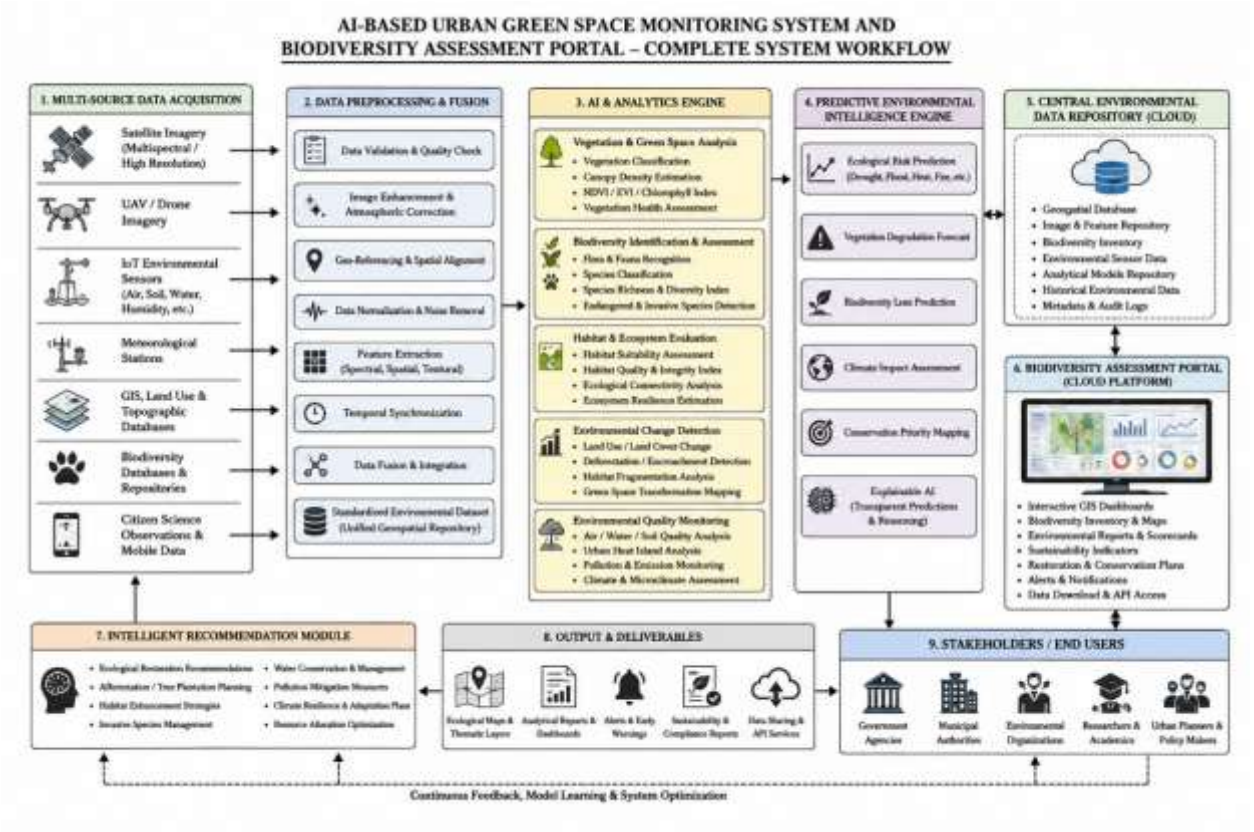


Figure 1: System Work Flow

Mrs. Saranya N | Dharnish J | Bharathwaj R | Sasidharan R

Name and Signature of the \\Applicants

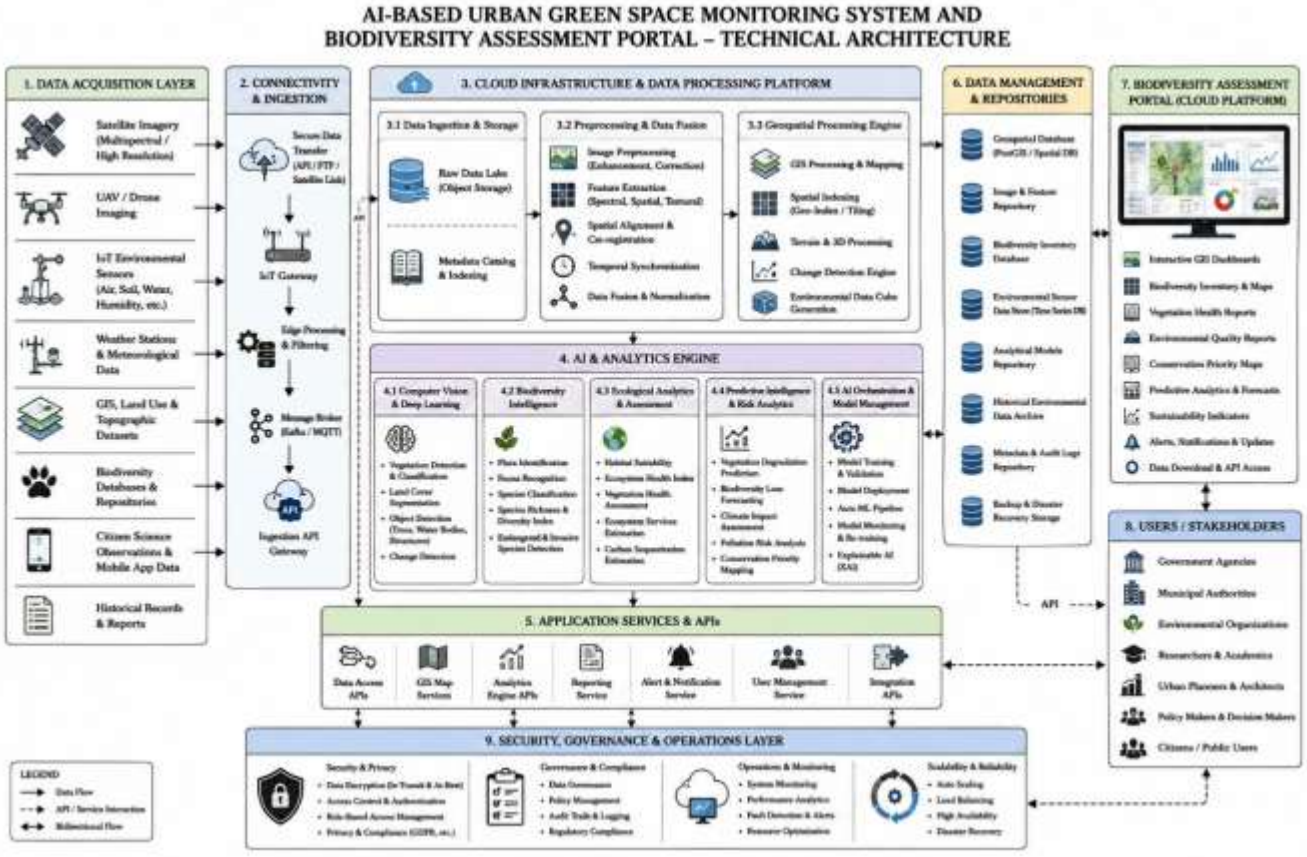


Figure 2: Technical Architecture

Mrs. Saranya N | Dharnish J | Bharathwaj R | Sasidharan R

Name and Signature of the Applicants

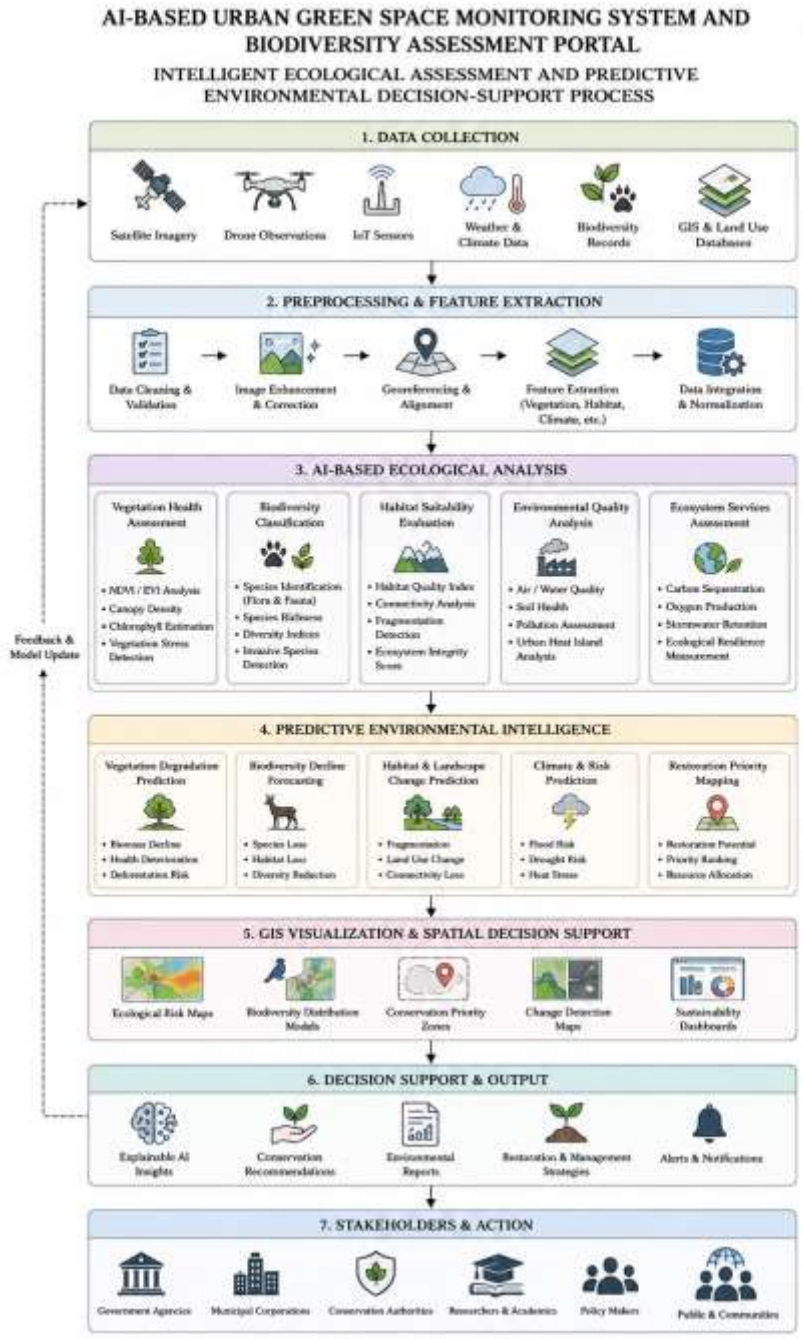


Figure 3: Intelligent Environmental Monitoring & Biodiversity Assessment Process Flow

Mrs. Saranya N	Dharnish J	Bharathwaj R	Sasidharan R

Name and Signature of the Applicants

FORM 1 THE PATENTS ACT 1970 (39 of 1970) and THE PATENTS RULES, 2003 **APPLICATION FOR GRANT OF PATENT** (See section 7, 54 and 135 and sub-rule (1) of rule 20)	**(FOR OFFICE USE ONLY)**

	Application No.	
	Filing date:	
	Amount of Fee paid:	
	CBR No:	
	Signature:	

1. APPLICANT"S REFERENCE / IDENTIFICATION NO. (AS ALLOTTED BY OFFICE)	

2. TYPE OF APPLICATION [Please tick (√) at the appropriate category]

Ordinary (√)		Convention ( )		PCT-NP ( )	
Divisional ( )	Patent of Addition ( )	Divisional ( )	Patent of Addition ( )	Divisional ( )	Patent of Addition ( )

3A. APPLICANT(S)

Name in Full	**Nationality**	**Country of Residence**	**Address of the Applicant**
Mrs. Saranya N	Indian	India	Assistant Professor, Department of Computer Science and Business Systems, Bannari Amman Institute of Technology, Sathyamangalam, Erode - 638401
Dharnish J	Indian	India	Student, Department of Computer Science and Business Systems, Bannari Amman Institute of Technology, Sathyamangalam, Erode – 638401
Bharathwaj R	Indian	India	Student, Department of Computer Science and Business Systems, Bannari Amman Institute of Technology, Sathyamangalam, Erode - 638401

Saidharan R	Indian	India	Student, Department of Computer Science and Business Systems, Bannari Amman Institute of Technology, Sathyamangalam, Erode - 638401

3B. CATEGORY OF APPLICANT [Please tick (√) at the appropriate category]

Natural Person (√)	Other than Natural Person			
	Small Entity ( )	Startup ( )	Others ( )	Educational Institution ( )

4. INVENTOR(S) [Please tick (√) at the appropriate category]

Are all the inventor(s) same as the Applicant(s) named above?	Yes (√)	No ( )

If "No", furnish the details of the inventor(s)

Name in Full	Nationality	Country of Residence	Address of the inventor
	Indian	India	

5. TITLE OF THE INVENTION: AI-BASED URBAN GREEN SPACE MONITORING SYSTEM AND BIODIVERSITY ASSESSMENT PORTAL INCORPORATING MULTI-SOURCE GEO-SPATIAL ANALYTICS, ECOLOGICAL INTELLIGENCE, AND PREDICTIVE ENVIRONMENTAL DECISION SUPPORT FRAMEWORK

6. AUTHORISED REGISTERED PATENT AGENT(S)	IN/PA	
	Name	
	Mobile No.	

7. ADDRESS FOR SERVICE OF APPLICANT IN INDIA	Name	Mrs. Saranya N
	Postal Address	Assistant Professor, Department of Computer Science and Business Systems, Bannari Amman Institute of Technology, Sathyamangalam, Erode - 638401
	Mobile No.	**+91 77088 25195**
	E-mail ID	**saranyanj@bitsathy.ac.in**

8. ~~IN CASE OF APPLICATION CLAIMING PRIORITY OF APPLICATION FILED IN CONVENTION COUNTRY, PARTICULARS OF CONVENTION APPLICATION~~

Country	Application Number	Filing date	Name of the Applicant	Title of the Invention	IPC (as classified in the convention country)

9. ~~IN CASE OF PCT NATIONAL PHASE APPLICATION, PARTICULARS OF INTERNATIONAL APPLICATION FILED UNDER PATENT CO-OPERATION TREATY (PCT)~~

International application number	International filing date

10. ~~IN CASE OF DIVISIONAL APPLICATION FILED UNDER SECTION 16, PARTICULARS OF ORIGINAL (FIRST) APPLICATION~~

Original (first) application No.	Date of filing of original (first) application

11. ~~IN CASE OF PATENT OF ADDITION FILED UNDER SECTION 54, PARTICULARS OF MAIN APPLICATION OR PATENT~~

Main application/patent No.	Date of filing of main application

12. DECLARATIONS

(i) Declaration by the inventor(s)

(In case the applicant is an assignee: the inventor(s) may sign herein below or the applicant may upload the assignment or enclose the assignment with this application for patent or send the assignment by post/electronic transmission duly authenticated within the prescribed period).

We, the above named inventors are the true & first inventors for this Invention and declare that the applicants herein are our assignee or legal representative.

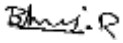
Dated This on 7th August 2026

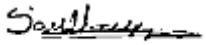
a) Signature (s) : 

b) Name(s): Mrs. Saranya N

a) Signature (s) : 

b) Name(s): Dharnish J

a) Signature (s) : 
b) Name(s): Bharathwaj R

a) Signature (s) : 
b) Name(s): Sasidharan R

(ii) Declaration by the applicant(s) in the convention country

(In case the applicant in India is different than the applicant in the convention country: the applicant in the convention country may sign herein below or applicant in India may upload the assignment from the applicant in the convention country or enclose the said assignment with this application for patent or send the assignment by post/electronic transmission duly authenticated within the prescribed period)

I/We, the applicant(s) in the convention country declare that the applicant(s) herein is/are my/our assignee or legal representative.

NA

(iii) Declaration by the applicant(s)

I/We the applicant(s) hereby declare(s) that: -

- ~~I am/~~ we are in possession of the above-mentioned invention.
- The Complete specification relating to the invention is filled with this application.
- ~~The invention as disclosed in the specification uses the biological material from India and the necessary permission from the competent authority shall be submitted by me/us before the grant of patent to me/us.~~
- ~~T~~here is no lawful ground of objection(s) to the grant of the Patent to me/us.
- ~~I am/we are the true & first inventor(s).~~
- ~~I am/~~we are the assignee or legal representative of true & first inventor(s).
- ~~The application or each of the applications, particulars of which are given in Paragraph 8, was the first application in convention country/countries in respect of my/our invention(s).~~
- ~~I/We claim the priority from the above mentioned application(s) filed in convention country/countries and state that no application for protection in respect of the invention had been made in a convention country before that date by me/us or by any person from which I/We derive the title.~~
- ~~My/our application in India is based on international application under Patent Cooperation Treaty (PCT) as mentioned in Paragraph 9.~~
- ~~The application is divided out of my /our application particulars of which is given in Paragraph 10 and pray that this application may be treated as deemed to have been filed on~~ ~~DD/MM/YYYY~~ ~~under section 16 of the Act.~~ ~~The said invention is an improvement in or modification of the invention particulars of which are given in Paragraph 11.~~

13. FOLLOWING ARE THE ATTACHMENTS WITH THE APPLICATION

Form 2

Item	**Details**	**Fee**	**Remarks**
Complete specification	No. of pages: 32		
No. of Claim(s)	No. of Pages: 3 and No. of claims: 6		
Abstract	No. of pages: 2		
No. of Drawing(s)	No. of drawing: 3 and No. of pages: 3		